

Test House WIP Analysis - Executive Summary

Date: July 6, 2026 **Analysis Period:** April - June 2026 (Q2) **Assumptions:**

- 5-day turnaround at test houses
 - 50% of orders send 100% of material; 50% send 5% sample only
-

Bottom Line: ~\$1.1 Million WIP

Approximately **\$1.1 million** in material value is at test houses on any given workday, representing **33 material orders** in steady-state processing (with varying sample sizes).

This estimate **excludes extreme outliers** (top/bottom 5%) and assumes half of orders send full material while half send only 5% samples for testing.

Key Findings

Material Flow

- **440 material orders** analyzed (excluding outliers)
- **6.67 orders sent per workday** to test houses
- **33 orders in-process** at any time (5-day float)
- **\$28.1M total material value** purchased in Q2 2026
- **Effective material at test house:** ~52.5% of order value (weighted avg)
 - 50% of orders: 100% of material sent

- 50% of orders: 5% sample sent

Average Material Order Value (Total Purchase)

- **Mean:** \$63,912 (trimmed)
- **Median:** \$23,055 (typical order)
- **P25-P75 Range:** \$4,980 - \$93,748

Weighted Average Material AT TEST HOUSE

- **Mean:** \$33,554 (52.5% of order value)
- **Median:** \$12,104 (52.5% of order value)

WIP Estimate Scenarios

Scenario	Calculation	WIP Value	Use Case
Trimmed Average (RECOMMENDED)	33.4 orders × \$33,554	\$1,119,021	Financial planning, typical operations
Trimmed Median	33.4 orders × \$12,104	\$403,664	Conservative estimate
Full Dataset Average (100% sent)	37.4 orders × \$93,201	\$3,481,055	If all material sent (unrealistic)

Recommendation: Use **\$1.1M** as baseline WIP estimate for cash flow and financial modeling.

What We Measured

CORRECT: Material Value at Test Houses

- Components/inventory physically at test houses

- Found by identifying material purchases on RFQs that also had testing service purchases
- 493 total material VQs on 179 RFQs (trimmed to 440 after removing outliers)

✗ **WRONG (Initial Analysis): Testing Service Charges**

- What test houses bill us for testing (avg \$2,575)
- This is the **cost of testing**, not the **value of what's being tested**
- Material orders are 36x larger than testing fees

Partial Sampling Model

Of the **33 orders in-process** at any given time (5-day turnaround):

Order Type	Count	Material Sent	Avg Value at Test House	Total WIP
Full material sent	~17 orders	100%	\$63,912	~\$1,066,000
Sample only	~17 orders	5%	\$3,196	~\$53,000
Weighted Total	33.4 orders	52.5% avg	\$33,554	~\$1,119,000

This assumes **50% of orders send full material** while **50% send only 5% samples**.

Data Quality & Methodology

Strengths

✓ Based on 440 actual transaction records (cost × qty from system) ✓ Trimmed dataset removes extreme outliers (top/bottom 5%) ✓ Logical linkage: material +

testing on same RFQ = material at test house  Conservative: only includes clearly paired material/service orders

Assumptions & Caveats

 **5-day turnaround assumed** - actual may vary by vendor/service type

 **Partial sampling assumption:**

- 50% of orders send 100% of material to test house
- 50% of orders send only 5% sample to test house
- Weighted average: 52.5% of material value at test house
- Actual split may vary by order type, vendor, or testing requirements

 **Trimmed data excludes:**

- 24 smallest orders (< \$4,980)
- 25 largest orders (> \$93,748)
- These outliers add \$17.8M to total but are atypical

 **May not capture all material:**

- Customer consignment material (not our purchase)
- Material sent without testing service on same RFQ
- Repeat testing without new VQ generation

Material Vendor Breakdown

85% of material comes from Global Sourcing vendors:

Vendor Type	Orders	Total Value	Avg Order
Global Sourcing	382	\$39.0M	\$102,156
New/Ungraded	13	\$2.7M	\$208,900

Vendor Type	Orders	Total Value	Avg Order
Quality	18	\$1.1M	\$59,082
Other	80	\$3.2M	varies

Key Insight: This is primarily overseas component purchases requiring testing/certification before resale.

Comparison of Analyses

Analysis Version	What Was Measured	Avg at Test House	Est. WIP (5-day)	Notes
Initial (WRONG)	Testing service charges	\$2,575	\$68K	Measured testing fees, not material
Corrected (All Data, 100%)	Material value	\$93,201	\$3.5M	Assumes all material sent
Trimmed (100% sent)	Material excl. outliers	\$63,912	\$2.1M	Assumes all material sent
Trimmed w/ Sampling (RECOMMENDED)	Material excl. outliers (52.5% sent)	\$33,554	\$1.1M	Reflects partial sampling

Recommendations

For Financial Planning

1. **Use \$1.1M as WIP baseline** for cash flow modeling
2. **Actual range: \$900K - \$1.3M** depending on order mix and sampling rates
3. **Flag orders over \$250K** for separate tracking (7% of volume, high \$ impact)
4. **Sensitivity to sampling assumption:**
 - If 100% of material sent: WIP = \$2.1M (91% higher)
 - If 75%/25% split (vs 50%/50%): WIP = \$1.7M
 - Current estimate assumes 50/50 split

For Operations

1. **Validate assumptions** by vendor:
 - Survey actual turnaround times (5-day assumption)
 - Track actual material sent vs sample-only orders
 - May vary by service type or vendor requirements
 - High-value orders may have different SLAs
2. **Track material separately from service charges:**
 - Service charges: ~\$68K WIP (testing fees, 5-day float)
 - Material value: ~\$1.1M WIP (components at test houses)
 - Total exposure: ~\$1.2M
3. **Consider inventory tracking improvements:**
 - Currently inferred from RFQ pairing
 - Could track explicitly via warehouse movements or locators
 - Track full vs sample shipments to refine 50/50 assumption

SQL Query Used

```

-- Material orders on service RFQs, excluding top/bottom 5%
WITH service_rfqs AS (
    SELECT DISTINCT chuboe_rfq_id
    FROM adempiere.chuboe_vq_line
    WHERE isactive = 'Y'
        AND ispurchased = 'Y'
        AND chuboe_vendortype_id = 1000006 -- Services
        AND created >= '2026-04-01' AND created < '2026-07-01'
        AND chuboe_rfq_id IS NOT NULL
),
material_orders AS (
    SELECT
        cost * qty as order_value,
        PERCENT_RANK() OVER (ORDER BY cost * qty) as percentile
    FROM adempiere.chuboe_vq_line
    WHERE isactive = 'Y'
        AND ispurchased = 'Y'
        AND chuboe_rfq_id IN (SELECT chuboe_rfq_id FROM service_rfqs)
        AND chuboe_vendortype_id != 1000006
        AND cost IS NOT NULL AND qty IS NOT NULL
        AND cost > 0 AND qty > 0
)
SELECT
    COUNT(*) as order_count,
    ROUND(AVG(order_value)::numeric, 2) as avg_value,
    ROUND(PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY order_value)) as median_value
FROM material_orders
WHERE percentile >= 0.05 AND percentile <= 0.95; -- Trim 0.05 and 0.95

```

Quick Reference Card

WIP at Test Houses (5-day turnaround, 50/50 sampling):

Total Orders In-Process:	33.4 orders
└ Full material (50%):	~17 orders × \$63,912 = \$1,066,488

└ Sample only (50%): ~17 orders × \$3,196 = \$53,000

Total Material WIP: \$1,119,000

Testing Service WIP: \$68,000

TOTAL EXPOSURE: ~\$1,190,000

Daily Flow:

- Orders sent per day: 6.67
- Effective material rate: 52.5% of order value
- Daily WIP churn: ~\$224K sent, ~\$224K returned

Sensitivity:

- If 100% material sent: WIP = \$2.1M (+91%)
- If 7-day turnaround: WIP = \$1.6M (+40%)
- If median order value: WIP = \$404K (-64%)

Files Generated

1. `test_house_wip_SUMMARY.md` - This executive summary (RECOMMENDED)
2. `test_house_wip_FINAL.md` - Comprehensive analysis with all data
3. `test_house_wip_executive_summary_REVISIED.md` - Initial corrected version
4. `service_house_vqs_3months.txt` - Service VQ counts by vendor/month

Prepared by: Analytics Team **Contact:** For questions about methodology or to validate assumptions